analogy, which threw the French prince for a loop. While de Broglie's reply was largely accurate, Pauli remained unsatisfied.

Another, more serious objection to de Broglie's interpretation came from Hans Kramers, a Dutch physicist who had been one of Bohr's students. He pointed out that when a photon bounces off of a mirror, the mirror must recoil slightly from the impact. But de Broglie's theory, according to Kramers, could not explain the mirror's recoil. De Broglie admitted he could not answer this question. Unbeknownst to de Broglie or Kramers, it was in fact possible to explain the mirror's recoil using de Broglie's theory—it merely required treating both the photon and the mirror as quantum objects, not just the photon. But de Broglie, like most other physicists at the time, thought that quantum physics applied only to small objects, and this left him unable to reply to Kramers. Soon after the conference, de Broglie himself gave up on his ideas, for reasons related to Kramers's objection.

Born and Heisenberg spoke next, presenting their matrix-based formulation of quantum physics, in which irreducibly random quantum jumps played a central role. Toward the end of their presentation, they boldly claimed that quantum physics was "a closed theory, whose fundamental physical and mathematical assumptions are no longer susceptible of any modification." In other words, quantum physics was done, fully cooked: there was no further need to dig into the innards and find something more, either mathematically or interpretationally. Later on, Bohr spoke, mostly rehashing his Como lecture, emphasizing that the wave and particle descriptions of quantum phenomena were complementary rather than contradictory: both necessary for a complete description, but never to be used to describe the same object at the same time.

Finally, after several days of sitting and listening while saying almost nothing, Einstein rose to speak during a period of open discussion. He had been passing notes with his close friend Paul Ehrenfest, gently mocking the Copenhagen camp, and had waited to carefully formulate his thoughts before he replied to them. Everyone in the room knew that Einstein had serious reservations about Bohr and Heisenberg's ideas. Now, all eyes were on him as he walked up to the chalkboard to sketch out a simple thought experiment that contained a devastating critique of the Copenhagen interpretation.